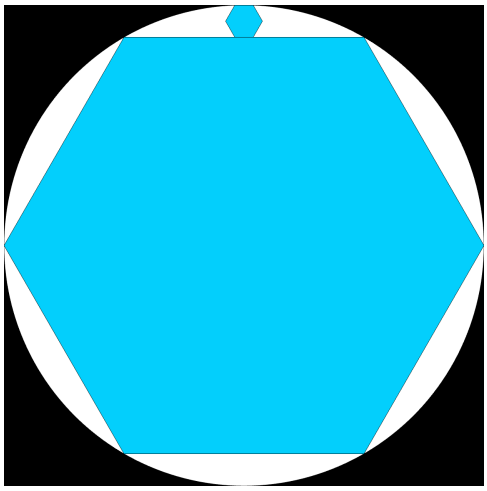
Classic Riddler

21 January 2022

Riddle:

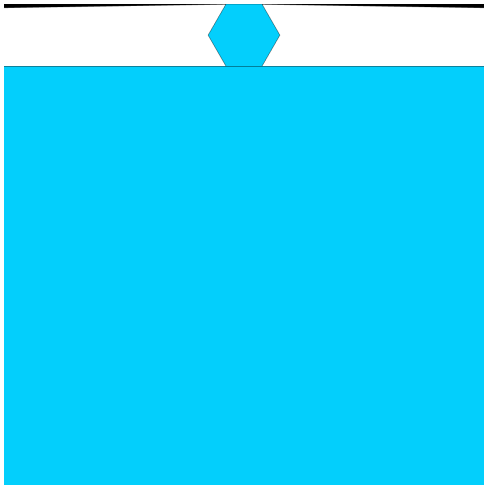
Rumor has it that readers of The Riddler Ellora Sarkar and Daniel Gomez are getting married this weekend. Congratulations to you both! In keeping with the hexagonal design of your wedding backdrop, I thought you might enjoy the following puzzle. (Riddler Nation, this one's for you, too!)

The larger regular hexagon in the diagram below has a side length of 1.



What is the side length of the smaller regular hexagon?

Extra credit: If you look very closely, there are two more, even smaller hexagons on top. Can you see them? No? Maybe this animation will help:

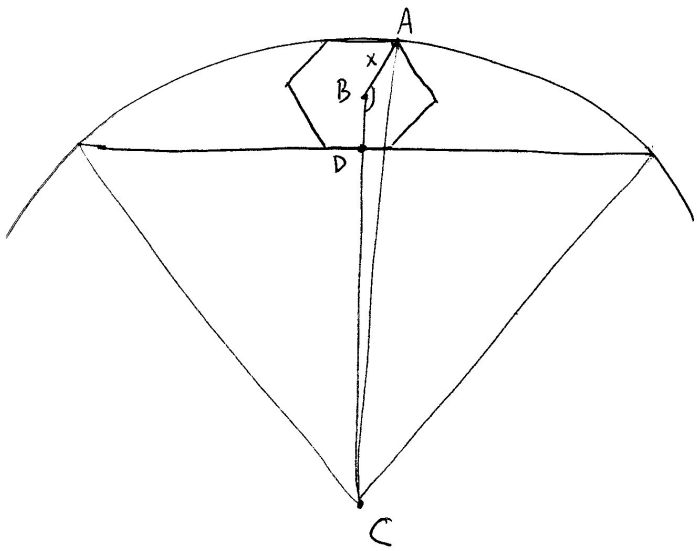


What are the side lengths of these two even smaller hexagons?

Solution:

Although it is not stated, I am assuming that the smaller hexagon is tangent to the larger hexagon, the larger hexagon's vertices lie on the circle, and the smaller hexagon's top two vertices lie on the circle. With this more

complete description, I have drawn a diagram with some points labeled. C is at the center of the circle (and the larger hexagon), and B is at the center of the smaller hexagon. In a regular hexagon, the side length is equal to the distance from the center to any vertex. Thus, if x is the length of $\overline{\mathrm{AB}}$, then x is also the side length of the smaller hexagon.



In the diagram, $\overline{\mathrm{AB}}$ is equal to 1. $\overline{\mathrm{CD}}$ is the altitude of a equilateral triangle of side length 1, which is $\sqrt{3}/2$. Similarly, $\overline{\mathrm{BD}}$ is $(\sqrt{3}/2)x$. The value of x can be solved using the law of cosines with the triangle $\triangle\mathrm{ABC}$:

$$1^2 = \left(\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}x\right)^2 + (x)^2 - 2\left(\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}x\right)(x)\cos(150^\circ)$$

Solving this equation conveniently leads to the exact solution $\boxed{\mathbf{1/13}}$.